

Design FMEA — VBF Power-Tool Cell (V4 Margin-fix)

Case: t3_r1_flash | Generation date: 2026-08-25 | Doc: VBF-T3R1FLASH-DFMEA-01

Qualitative version, based on simulation signals (severity S / occurrence O: high/medium/low; RPN = qualitative S×O matrix).

Failure mode	Failure cause	Simulation signal (detectability basis)	S	O	RPN	Design-side mitigation
Lithium plating at 4C fast charge	Negative electrode surface potential < 0 V (anode fills, overpotential grows)	r2_v4_4c_dfn.json anode_potential_v min +22.6 mV (baseline -0.19 V; V1 -3.9 mV)	high	low (margin 22.6 mV)	high×low=medium	Thicker anode (N/P ¹), ε _{neg} 0.38, r _{neg} 1.0 μm, high σ _{e/t} ■, 45 °C charge temp
Thermal runaway risk from fast-charge heat	Heat generation > cooling at 4C	T_max 321.94 K vs 333.15 K limit (baseline 354.3 K)	high	low	medium	Low-DCR thin-electrode design + forced-air cooling h=100 W/m²/K
Insufficient capacity	Electrode thinning cuts capacity	3.438 Ah vs ≥2.0 Ah (baseline 4.95 Ah)	medium	low	medium	Thickness chosen to keep >3 Ah margin
5C rate capability loss	Slow solid diffusion (D_pos=4e-15 m²/s) limits high-rate utilization	retention 0.987 vs ≥0.95 (baseline 0.087)	high	low	medium	Small particles r_pos 0.8 μm (τ ~160 s), thin electrodes, high-σ electrolyte
Electrolyte oxidative decomposition	Cell voltage above electrolyte stability window	Not computed in this case (molecular funnel skipped, start_stage=3; final DFT endorsement caliber not run, real_compute=false)	medium	low	low	Voltage window 2.5–4.2 V within standard EC/EMC LiPF6 stability (literature); honest note: not re-verified at molecular scale

Conclusion

Highest-risk items (plating, thermal, rate capability) all show large simulation margins in the final design (plating +22.6 mV, T -11.2 K, retention +3.7 pts) and have been mitigated in the design as implemented. Complete FMEA including process/supplier failure modes: N/A (beyond pure simulation boundary).